$\therefore$ Volume of the sphere with radius 'r'

$$= \frac{4}{3}\pi r^3$$

$$= \frac{4}{3}\pi \left(\frac{a}{2}\right)^3$$

$$= \frac{4}{3}\pi \left(\frac{a^3}{8}\right)$$

$$= \frac{\pi a^3}{6} \quad \ldots (1)$$

In a simple cubic arrangement, number of spheres belongs to a unit cell is equal to one

$$\therefore \text{Total volume occupied by the spheres in sc unit cell} = 1 \times \left(\frac{\pi a^3}{6}\right) \quad \ldots (2)$$

Dividing (2) by (3)

$$\text{Packing fraction} = \frac{\left(\frac{\pi a^3}{6}\right)}{\left(a^3\right)} \times 100 = \frac{100\,\pi}{6}$$

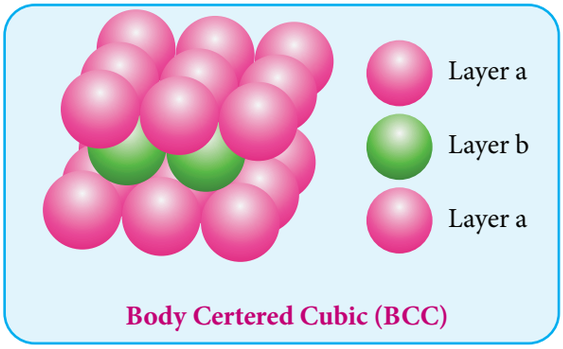
$$= 52.38\%$$

i.e., only 52.38% of the available volume is occupied by the spheres in simple cubic packing, making inefficient use of available space and hence minimizing the attractive forces.

More to know Of all the metals in the periodic table, only polonium crystallizes in simple cubic pattern.

6.6.4 Body centered cubic arrangement

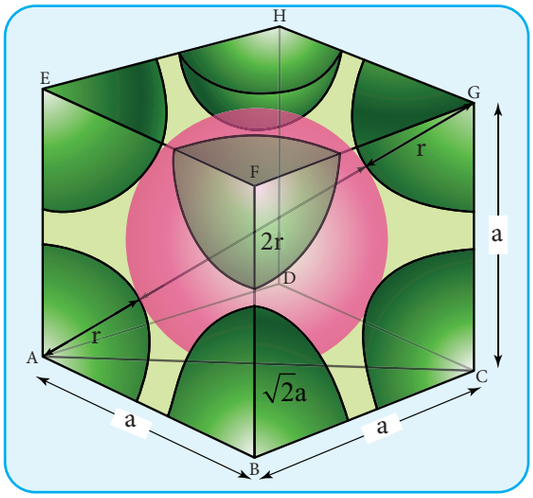
In this arrangement, the spheres in the first layer (A type) are slightly separated and the second layer is formed by arranging the spheres in the depressions between the spheres in layer A as shown in figure. The third layer is a repeat of the first. This pattern ABABAB is repeated throughout the crystal. In this arrangement, each sphere has a coordination number of 8, four neighbors in the layer above and four in the layer below.



Body Centered Cubic (BCC)

Packing efficiency:

Here, the spheres are touching along the leading diagonal of the cube as shown in the fig.



In ΔABC

$$AC^2 = AB^2 + BC^2$$

$$AC = \sqrt{AB^2 + BC^2}$$

$$AC = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}\ a$$

In ΔACG

$$AG^2 = AC^2 + CG^2$$

$$AG = \sqrt{AC^2 + CG^2}$$

$$AG = \sqrt{\left(\sqrt{2}a\right)^2 + a^2}$$

$$AG = \sqrt{2a^2 + a^2} = \sqrt{3a^2}$$

$$AG = \sqrt{3}\ a$$

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